DESIGN AND NUMERICAL ANALYSIS OF A SUB-SURFACE TWO-BODY WAVE ENERGY CONVERTOR

Virag Mishra, Michael Kelly, Joseph Piacenza, and Solomon C. Yim

***Abstract*—Recent developments in self-reacting oscillating Wave Energy Convertors (WECs) point toward two popular design concepts. These designs are either semi-submerged system in which a surface piercing float reacts against a relatively still heave plate or a single body system that reacts against the ocean floor. The first design employs a rigid spar connection between the actuator and the reaction body; however, the second design uses a flexible belt connection between the actuator and the foundation. In this work, a two-body sub-surface WEC is proposed in which the submerged float reacts against a base that is actively dynamic. The dynamics of the reacting body is influenced by both the buoyancy parameters and the PTO generated tether tension. As a result, the two-body system oscillates either out-of-phase or in an antipodal mode. The optimal hydrostatic conditions that result in a design that can generate mechanical power, in single degree of freedom motion, under the wave excitation are discussed in this work. Multiple design candidates are evaluated under different wave conditions and exhaustive sensitivity analyses are carried out on various design parameters. The hydrodynamic analysis is carried out for different wave conditions observed off the coast of US. Optimal PTO parameters are determined for various designs. The analysis is carried out using "Ansys-Aqwa®".**

***Index Terms*—Hydrodynamics, Power Take Off (PTO), Wave Energy Converter (WEC)**

I. Introduction

DUE to the concerted effort to achieve energy independence as well as to reduce greenhouse gas emission, most of the development to date in offshore renewable energy systems have been targeted at "utility-scale" or "community-scale" devices. These devices have electricity generation capacities in the order of MW and kW respectively. However, another application area that is attracting investment in wave energy converters is "small-scale" devices. These devices can provide autonomous power to various ocean-bound systems, such as autonomous underwater vehicles [1], observation systems [2] and even aquaculture farms [3]. The expectation is that these devices can be used to power the on-site stand-alone systems which are located remotely, resulting in exponential growth in the sector often referred as "Blue economy" [4], [5].

II. Background

Numerical benchmarking of eight WECs with different working principles was conducted to evaluate their power matrices and site specific annual energy production [6]. Out of these, two operated close to seabed and only one was submerged near the sea surface. This shows that the focus of marine energy research has mainly been around floating devices. Some of the most studied floating WECs with heave plate are MBARI-WEC [7], OPT Powerbuoy, OSU-Lupa (Laboratory Upgrade Point Absorber) [8] and Wavebob [9].

Though the wave energy decays exponentially with depth in deep water regime, subsurface WECs provide the potential benefit of survivability and covertness. Possibly these are reasons why subsurface WECs are emerging as a preferred design choice for WEC developers. Some of the popular designs that are either bottom fixed or do not have a heave plate as reacting bodies are AWS Ocean Energy's Archimedes Waveswing, Symphony Wave Power's WEC, Carnegie Clean Energy's CETO 6 [10], and CalWave's xWave. CETO6 and xWave are multi tethered, multi DOF point absorbers which operate close to the surface. The Waveswing is an air pressure differential point absorber, tethered to the sea bed and Symphony Wave Power's WEC is a bottom-fixed device operating in shallow waters. Conceptual submerged WEC capable of producing 10 W of power with a float and six trubines has been proposed in [11].

In all the buoyant designs mentioned above the buoyancy of the float acts as the actuation force and either the foundation in the sea bed or the heave plate acts as a reaction body. In this work we propose a two-body submerged system with a float and heave plate connected by a tether and the entire system is moored to the foundation. The float will act as an actuator and the buoyancy of the heave plate will aid to power generation. This work will evaluate the performance for the above described WEC and compare six different design candidates at different wave energy sites in the US.

This paper provides an overview of the initial design process of such a device. Previous studies comparing the prevalence of submerged and small-scale WECs with other WEC configurations are discussed in Sec-

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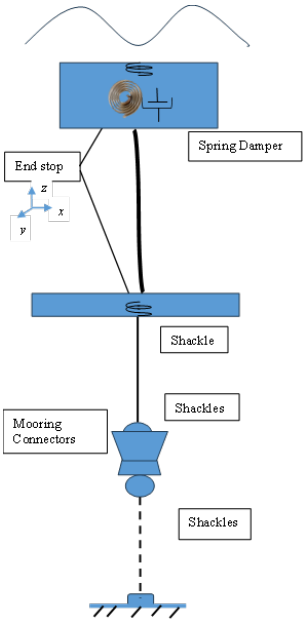


Figure 1. Sketch of the two-body sub-surface WEC under analysis

tion II, the operating principle and primary dimension of the device is derived and discussed in Section III. The numerical modelling approach is described in Section IV. The details of the deployment sites are discussed in Section V. The results and the following discussion are presented in Section VII.

III. Device Details

The WEC design under study is a two-body point absorber consisting of a submerged float and a heave plate. The design of the WEC supports the sub-surface operation. Sketch of the primary components used in the numerical modelling are shown in Fig. 1. The float and heave plate are connected by a tether which is attached to the top of the heave plate and the bottom of the float. The heave plate is moored to the sea bed using a mooring line. The mooring line is attached to the bottom of the heave plate and the other end is attached to a foundation on the sea bed. The tether and mooring line are assumed to be linear elastic with a stiffness of k and $0.1k$ respectively. The tether and mooring line are assumed to be massless and inextensible.

To determine optimal device configuration multiple designs were analyzed. In this paper preliminary results for some of the designs considered are discussed and compared. The relative dimensions of the float and the heave plate for various designs considered are shown in Fig. 2. A summary of the important dimensions and characteristics of these designs are presented in Table I. R_f and R_{hp} are the radius of float and heave plate respectively and H_f, H_{hp} are the height of float and heave plate. Similarly, W_f and W_{hp} are the weight of the float and heave plate. The primary difference between various designs is that the total weight of the design D0 was 8000 kg, however other designs are comparatively lighter in the range 2100 kg (D1, D2), 2500 kg (D3, D4) and 2966 kg (D5, D6). The tether tension in the design D0 was 23696 N, which would increase the CAPEX, as the cost is directly proportional to the tether cost. However, the tether tension in other designs is significantly lower.

IV. Numerical Modelling

Commercially available software "Ansys-Aqwa®" was used to both calculate the frequency dependent parameters and to simulate the time-dependent dynamic behavior of the WEC. Each simulation was run for 1800 s with a time step of 0.25 s. The first 100 s of the simulation was considered as the wave ramp-up period and discarded, however, the remaining data was used to calculate the average power produced by the WEC. The average power was calculated by integrating the instantaneous power between time steps and dividing by the fraction of an hour between them using (1)

$$E_{\text{avg}} = \frac{t_2 - t_1}{3600} \int_{t_1}^{t_2} P_{pto}(t)\, dt \tag{1}$$

where $P_{pto}(t)$ is the instantaneous power from the Power Take-Off system and $t_1 = 100$ s and $t_2 = 1800$ s are the start and end times of the integration, respectively.

A. Generalized WEC modeling

The Cummins equation formulation [12] is used by the time-domain model in Aqwa to solve the general force balance equation for a WEC shown in (2).

$$\begin{aligned}(\mathbf{M} + \mathbf{A}_\infty)\,\ddot{\mathbf{X}}(t) + \int_0^t \mathbf{K}(t-\tau)\dot{\mathbf{X}}(\tau)\,d\tau \\ + \mathbf{C}_{HS}\mathbf{X}(t) = \mathbf{F}_{ext}(t)\end{aligned} \tag{2}$$

In the above equation, the terms are defined as follows: $\mathbf{M}$ is the physical mass and inertia matrix of the structure. $\mathbf{A}_\infty$ is the added mass at infinite frequency. AQWA derives this from its initial frequency-domain potential-flow solution [13]. The term $\int_0^t \mathbf{K}(t-\tau)\dot{\mathbf{X}}(\tau)\,d\tau$ is the radiation convolution integral, which represents the fluid memory effects caused by wave radiation damping. The radiation impulse response function (RIRF), $\mathbf{K}(t)$, is calculated from the frequency-domain damping coefficients $\mathbf{B}(\omega)$ as

$$\mathbf{K}(t) = \frac{2}{\pi}\int_0^\infty \mathbf{B}(\omega)\cos(\omega t)\,\mathrm{d}\omega. \tag{3}$$

$\mathbf{C}_{HS}$ is the hydrostatic stiffness matrix. The external force vector $\mathbf{F}_{ext}(t)$ includes excitation forces, mooring-line forces, fender forces, Power Take-Off forces, and nonlinear viscous drag.

Though Aqwa solves this equation by directly integrating the time-consuming convolution term numerically, some other solvers can convert the convolution term into an equivalent state-space model to massively speed up computation times without losing accuracy when compared to Aqwa [14].

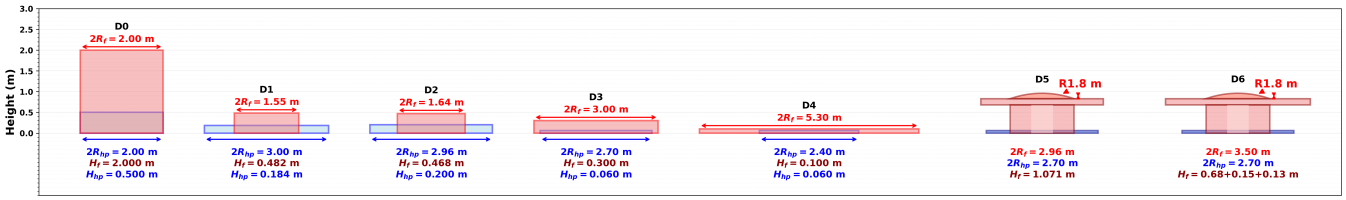


Figure 2. Representative dimensions and designs of the float and heave plate for hydrodynamic analysis. The heave plate and float were connected to the tether not shown in the above image.

Table I
DESIGN PARAMETERS AND LOADING CHARACTERISTICS OF THE CANDIDATE WEC CONFIGURATIONS.

Design	W_f (kg)	W_{hp} (kg)	$W_f + W_{hp}$ (kg)	R_f (m)	R_{hp} (m)	H_f (m)	H_{hp} (m)	Tether (N)	Mooring (N)
D0	4000	4000	8000	1	1	2	0.5	23931	201
D1	700	1400	2100	0.777	1.5	0.482	0.184	2325	1601
D2	800	1600	2400	0.82	1.48	0.468	0.2	2092	172
D3	1500	1000	2500	1.5	1.35	0.3	0.06	6606	203
D4	1500	1000	2500	2.65	1.2	0.1	0.06	7466	358
D5	2066	900	2966	1.48 (max)	1.35	1.027 (total)	0.06	5782	289
D6	2066	900	2966	1.75 (max)	1.35	0.853 (total)	0.06	5782	277

B. Cable modelling

A linear elastic cable module in Aqwa is the simplest numerical model of an actual mooring line. Aqwa requires the user to specify the unstretched length instead of pretension for elastic tethers. With the knowledge of stiffness and pretension (calculated as the difference between the buoyancy and the weight of the float and Heave Plate), the stretched length L^t of the tether and mooring can be determined using (4) and (6).

$$L^t = \frac{(V_f - V_f^{disp})\rho_w g}{k} \tag{4}$$

$$L_0^t = D_{hp} - (H_f + D_f) - L^t \tag{5}$$

$$L^m = \frac{(V_f + V_{hp}) - (V_f^{disp} + V_{hp}^{disp})\rho_w g}{0.1k} \tag{6}$$

$$L_0^m = D - (D_{hp} + H_{hp}) - L^m \tag{7}$$

The initial unstretched length L_0 of the tether and mooring can be calculated using (5) and (7) respectively. V_x^{disp} is the displaced volume of the respective cylindrical geometry (x can be either "f" for float or "hp" for heave plate.), ρ_w is the density of water, g is the acceleration due to gravity, k is the stiffness of the tether, D_x is the distance from the free surface to the respective cylindrical geometry, H_x is the height of the respective cylindrical geometry. D is the depth of water at deployment location (considered $= 70$ m for all the simulations).

C. PTO algorithm

The PTO system is modelled as a linear spring-damper system. The total force acting on the PTO system is the sum of the mechanical spring force and the generator force. This force is given by 8

$$F_{PTO} = \begin{cases} C + K_{mech}\Delta L^t + B\dot{\Delta L^t}, & \text{if } L^t > L_0^t \\ 0, & \text{if } L^t \leq L_0^t \end{cases} \tag{8}$$

where C is the constant pretension, K_{mech} is the spring stiffness, $\Delta L^t = (L^t - L_0^t)$ is the tether extension and B is the damping coefficient.

In this project two PTO models have been developed. These models are

1) Cable-Fender: In this model, the PTO is modelled as a combination of a linear elastic cable and a fender. Cable and fender are native elements in Aqwa and are connected in series. The spring stiffness is modeled using the cable element. A fender in Aqwa can have nonlinear stiffness, friction and damping. However, for this modelling only the damping properties of the fender were considered. Just like the cable element, the fender acts only in extension between a point on float and the heave plate.

2) Spring-Damper: A User Defined Function (UDF) is an external program that can be called from Aqwa and coupled with the solver to enhance the existing functionality. In this work the UDF is utilized to replace the numerical PTO modeling component of Aqwa (Cable-Fender) discussed above. This architecture will allow us to implement arbitrary force models like providing limits and providing component characteristics in future that cannot be defined in standard Aqwa input files.

For each design and every sea state, 58 combinations of stiffness and damping coefficients were used in the simulations. Fig. 3 shows the various combinations of PTO parameters considered in this analysis.

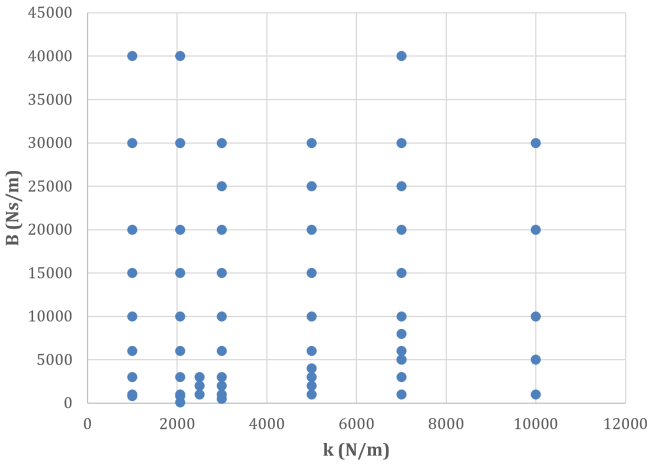


Figure 3. Various combinations of PTO parameters considered in the analysis.

V. Wave Climate at Deployment Locations

Based on the works of [15] on wave resource assessment off the coast of US Northwest, a set of guidelines were developed and included in a technical specification for wave resource assessment by International Electrotechnical Commission (IEC) [16]. The IEC brings together international marine energy experts to develop consensus-based standards that guide the development of a broad range of sectors. The current industry consensus is determination of representative sea states based on the energy weighted joint probability distribution function of primarily H_s and T_e/T_P [17].

Current approaches neither develop a multidimensional probabilistic description of the deployment site nor account for energy-weighted probabilities, and they typically preclude the evaluation of WEC performance across multiple deployment climates. A need for developing a methodology that could account for multiple deployment locations was presented in a Wave Energy Prize competition by US Department of Energy in 2016. A new methodology was proposed in this competition by [18] that applied a cluster analysis to scatter diagrams in order to determine a subset of sea states that could be scaled to find the average annual power flux at each wave climate. A common centroid representing of H_s and T_P was determined for the clusters. This procedure opens the possibility of using a single set of cluster centroid for different locations. Since WEC performance assessment simulations are computationally resource intensive, in this work the methodology described above is used to run simulations for a single set of cluster centroids and the power produced for each centroid is scaled depending on the site under consideration.

A set of 18 sea states, shown in Fig. 4, was used throughout the analysis in this report. Wave elevations were generated using the Bretschneider spectrum. Although these wave datasets do not all correspond to the same water depth (ranges from 42 m to 149 m), they provide a useful indication of WEC performance across varied wave profiles, since the same set of sea states is applied at each location shown in Fig. 5. Note that when the weights associated with the 18 sea states are used, they account for varying percentages of the total joint probability distribution (67%−95%), depending on location. This is because, at some locations, a significant portion of the year falls into a sea states not represented within any of the 18 sea states considered.

Table II
NDBC Buoys Chosen for Analysis

Buoy ID*	US Location	Depth [m]	Year
51207	Oahu, HI	81	2019
42001	Mid Gulf, New Orleans, LA	32	2018
46050	PACWAVE, OR	149	2017
46254	San Diego, CA	46	2019
41009	Canaveral, FL	42	2011
44018	Cape Cod, MA	44	2010

* All wave data comes from National Data Buoy Center (NDBC)

Hs/Tp	5.5	7.5	9.5	11.5	13.5	15.5
5.5		✓	✓	✓	✓	
4.5		✓	✓	✓	✓	
3.5		✓	✓	✓	✓	
2.5		✓	✓	✓	✓	
1.5		✓	✓	✓	✓	
0.75	✓	✓	✓	✓	✓	✓

Figure 4. The sea states used for the analysis of WEC performance

Site specific optimal PTO coefficient for the WEC are determined for each location shown in Fig. 5. These sites either host a testing facility or a deployment berth and are probable sites for the development of WEC farms. The significant wave height reported by NDBC (and used here) is the average of the highest one-third of all the wave heights during a 20-minute sampling period. The peak period is the period with maximum energy in the spectrum. The list of the buoy locations chosen for the analysis is presented in Table II.



Figure 5. Location of the NDBC buoys used for the analysis of wave climate

Occurrence data reported by NDBC buoys were used to calculate the annual average power at that location. A cluster analysis was applied to wave scatter diagrams to identify a representative subset of sea states that can be scaled to estimate the average annual power (AAP) flux. The subset is determined by minimizing the distance between each data point and its corresponding cluster centroid. Four pdist2 options for minimizing the distance function were compared these are Squared Euclidean, Standardized Euclidean, Eclidean

and Mahalanobis. This clustering is performed using the k-means algorithm implemented in the MATLAB Statistics Toolbox, where the number of clusters and the centroid values is specified by the user. The algorithm employs an iterative partitioning procedure that optimizes the associated memberships by minimizing the sum of point-to-centroid distances across all k clusters. As a result, each sea state is assigned to the cluster whose distance to centroid yields the minimum value, providing an optimally representative classification of the wave climate. In this paper results corresponding to "Mahalanobis" method are presented.

VI. Sensitivity Analysis

The sensitivity study summarized in Table III investigates the influence of mass ratio MR, float draft FD, and the distance between float–heave plate (Gap) on the hydrodynamic performance of the device. The parameters that were varied are shown in bold. The baseline configuration, D0, has a mass ratio of 1.0, float draft of 5 m, HP draft of 30 m and a float–HP gap of 23 m. To assess the effect of mass ratio, the float geometry and submergence were held constant while the mass ratio varied from 0.5 to 1.25. The influence of float draft was then examined by varying the draft from 2 m to 7 m, with the HP draft adjusted accordingly to maintain a constant 23 m float–HP gap. An additional design, D3, with a mass ratio of 1.5, was used to further investigate the effects of reduced float draft and varying float–HP gap. For this design, the two float draft that were studied are from 1 m and 2 m, while the HP draft was modified to preserve a 24.7 m gap. Finally, the float–HP gap was independently varied from 10 m to 24.7 m by changing the HP draft while keeping the float draft constant at 2 m. In addition to the above parameters the effect of varing the heave plate mass was also studied. The heave plate mass was varied from 900 kg to 800 kg and the corresponding mooring line pretension and unstretched length were calculated using (6) and (7). The parameters for this analysis are summarized in Table IV. The water depth was maintained at 70 m, ensuring that the observed changes in response are attributable solely to the selected sensitivity parameters.

Table III
Parameters for sensitivity analysis using design D0 and D3

Design	Sensitivity Parameter	Mass Ratio	FD (m)	HP Draft (m)	Gap (m)
D0	MR	**1.00**	5	30	23
D0	MR	**0.5**	5	30	23
D0	MR	**0.75**	5	30	23
D0	MR	**1.25**	5	30	23
D0	FD	1	**2**	27	23
D0	FD	1	**3**	28	23
D0	FD	1	**5**	30	23
D0	FD	1	**7**	32	23
D3	FD	1.5	**1**	26	24.7
D3	FD	1.5	**2**	27	24.7
D3	Gap	1.5	2	12.3	**10**
D3	Gap	1.5	2	17.3	**15**
D3	Gap	1.5	2	27	**24.7**

* MR-Mass Ratio, FD-Float Draft, Gap-Float-HP Gap

Table IV
Parameters for sensitivity analysis using design D5

HP (kg)	Equi. Vol Disp (m^3)	Mooring Pretension (N)	Unstretched length (m)
900	0.878	289	42.36
800	0.780	1270	40.4

VII. Results and Discussion

A. Verification of PTO models

The two PTO models, Cable-Fender and Spring-Damper that were discussed in Section IV-C, were compared for a single design (D0) under regular and irregular wave conditions. The comparison was made by analyzing the time series of velocity of the float, heave plate and the relative velocity between the two. The time series of velocity of the float, heave plate and the relative velocity between the two are shown for designs D0 and D3 in Fig. 6 and Fig. 7 respectively. Regular wave input is used in Fig. 6 and irregular wave excitation in Fig. 7. From these plots, it can be said that the results of important parameters obtained from UDF compare well with those obtained from the default solver. Since the UDF is a more flexible and robust model, it is used for all the subsequent simulations.

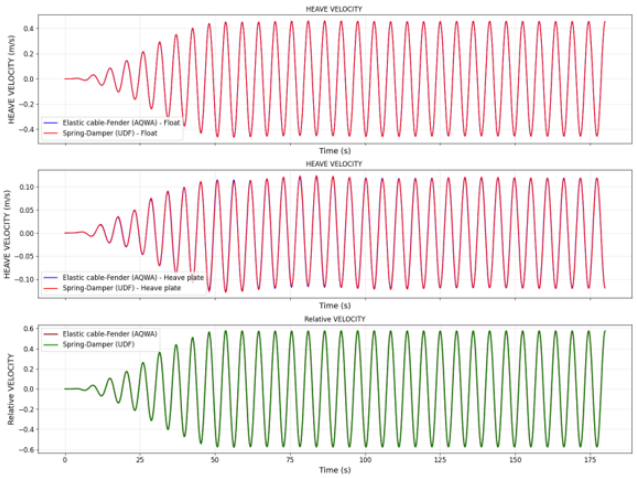


Figure 6. Variation in velocity of float, heave plate and the relative velocity between them for design D0 under regular waves of $H = 0.5$ m and $T = 5.5$ s

B. Variation of mass ratio

In this sensitivity test the mass of the float and heave plate was varied to study the effect of ratio of mass $\mathrm{MR} = \mathrm{M_F}/\mathrm{M_{HP}}$ on the average power absorbed by the wave energy converter for a PTO configuration with stiffness $k = 3000$ N/m and damping $C = 2000$ Ns/m. The variation of the power with H_s and T_p is shown in the Fig. 8. The power increases almost linearly with significant wave height, with progressively steeper slopes at lower mass ratios. At every wave height,

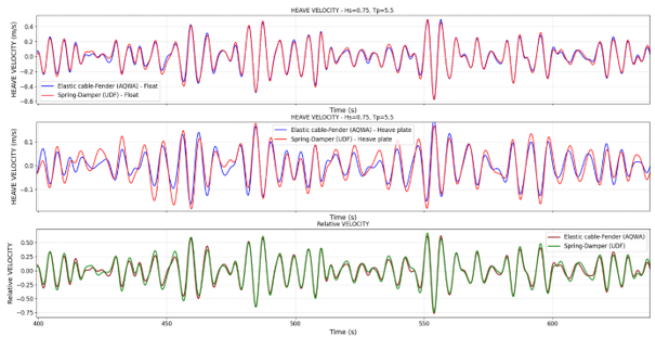


Figure 7. Variation in velocity of float, heave plate and the relative velocity between them for design D0 under irregular waves of $H_s = 0.75$ m and $T_p = 5.5$ s

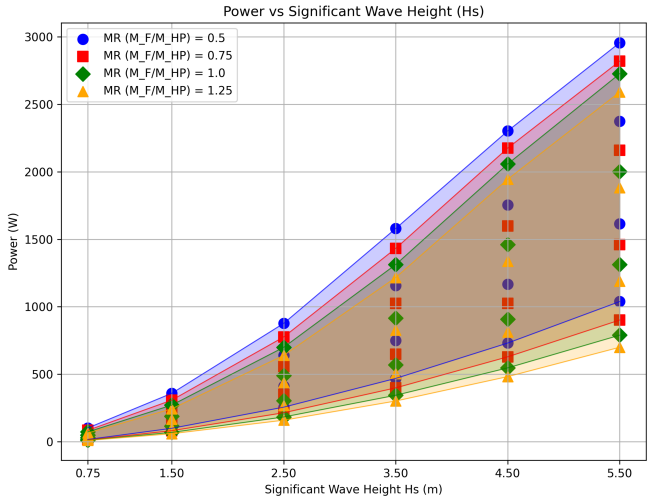
the ordering of the curves remains unchanged, indicating that increasing the float-to-heave plate mass ratio reduces the energy captured. Similarly, the second subplot shows that the average power decreases approximately linearly as the peak period increases from 7.5 s to 13.5 s. The reduction is substantial across all mass ratios, highlighting that the selected PTO configuration is better tuned to shorter-period waves. The separation between the curves persists throughout the period range, indicating that the effect of mass ratio is largely independent of wave period. This study demonstrates that themass ratio acts as an important design parameter that consistently shifts the power response, with lower mass ratios providing superior performance under all examined sea states. These results suggest that reducing the float-to-heave plate mass ratio (lighter float relative to the heave plate) improves the dynamic response of the system, allows the system to respond more effectively to wave excitation and enhances energy absorption without altering the overall dependence on the incident wave conditions.

C. *Variation of float draft*

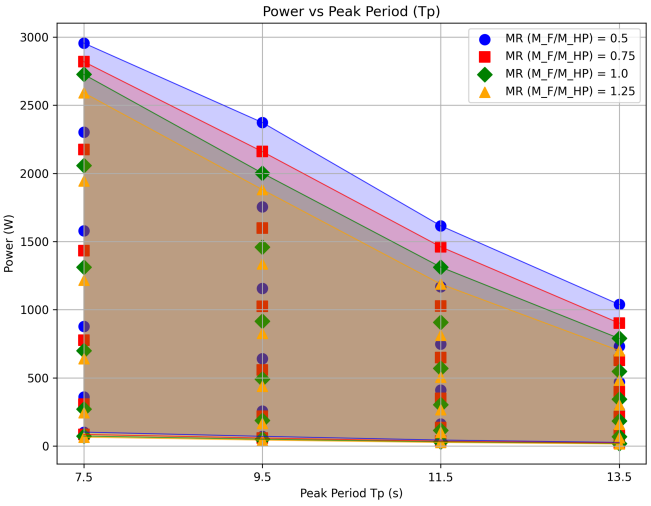
In this sensitivity test the draft of the float was varied to study the effect of float draft on the average power absorbed by the wave energy converter for a PTO configuration with stiffness $k = 3000$ N/m and damping $C = 2000$ Ns/m. The variation of the power with H_s and T_p is shown in Fig. 9. The float draft that were considered in this study are 2 m, 3 m, 5 m, and 7 m. The average power decreases substantially with increasing float draft, with a reduction of approximately 50% when the draft is increased from 2 m to 7 m. This trend is consistent across all examined sea statese. The results suggest that optimizing the float draft is critical for maximizing energy absorption and storm survival. On one hand shallower drafts increse the power capture and on the other hand deeper drafts reduce the dynamic response of the system protecting the critical mechanical components.

D. *Variation of initial float-heave plate gap*

The change in the average power when the float-heave plate gaps (Gap) was reduced from 25 m to 15 m and 10 m across all the sea states was studied for two PTO parameters $k = 7000$ N/m, $B =$



(a) Variation with significant wave height



(b) Variation with peak wave period

Figure 8. Variation in average power with mass ratio across the examined sea states.

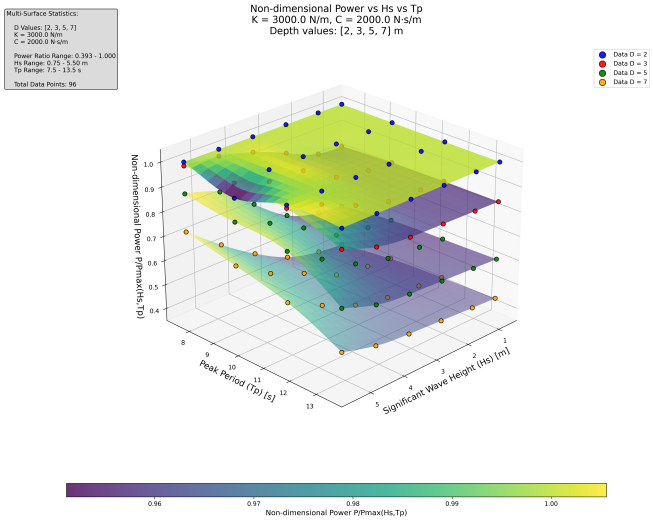


Figure 9. Substantial decrease in nondimensionalized average power output with increase in float draft, ($k = 3000$ N/m, $B = 2000$ Ns/m) across a range of sea states.

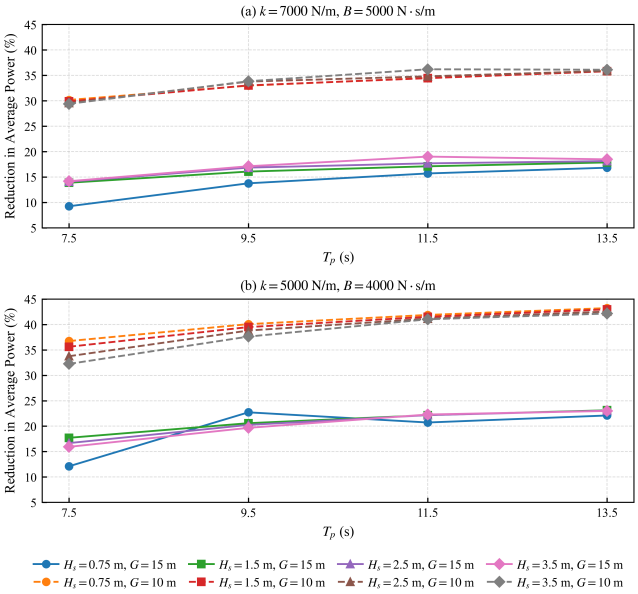


Figure 10. Reduction in the average power output of the wave energy converter with decrease in float–heave plate gaps (Gap) from 25 m to 15 m and 10 m across a range of sea states for two PTO configurations, ($k = 7000$ N/m, $B = 5000$ Ns/m and $k = 5000$ N/m, $B = 4000$ Ns/m).

5000 Ns/m and $k = 5000$ N/m, $B = 4000$ Ns/m. For both PTO configurations, decreasing the gap consistently reduced the average power output under all wave conditions. The relative reduction ranges from $9.3\% - 19.0\%$ and $12.1\% - 23.4\%$ for a Gap reduction from 25 m to 15 m. Similarly for a Gap reduction from 25 m to 10 m, the relative reduction ranges from $15.5\% - 30.0\%$ and $19.3\% - 36.0\%$, respectively for the two PTO configurations. It was also observed that the lower stiffness and damping configuration is more sensitive to the float–heave plate separation. The percentage loss generally increases with increasing wave period and reaches its maximum for long-period waves, where T_p ranges from 13.5 s to 15.5 s, though only a modest dependence on significant wave height was observed. The results demonstrate that increasing the float–heave plate separation enhances energy capture across all examined operating conditions, with the greatest relative benefit occurring under longer-period wave conditions. Results also suggest that optimizing the gap is particularly beneficial for PTO systems with lower stiffness and damping.

E. Variation of the heave-plate mass

For design D5, two variations of HP geometry were analyzed, to compare the effect of the weight of HP geometry in the average power. First case had heave plate mass of 900 kg and second had a mass of 800 kg. The results show that the change in heave plate mass affects the average power output. This change led to a change in various static properties tabulated in Table IV. The values of avg power obtained are shown in Fig. 11. The power obtained using 900 kg heave plate is 0.3% to 3% higher than 800 kg heave plate at various sea states. A heavier HP has higher equivalent displaced

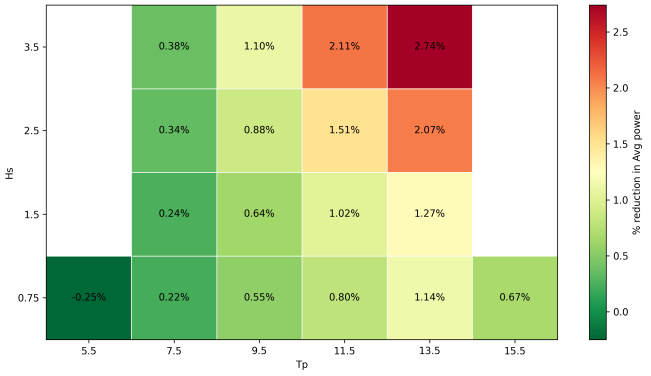


Figure 11. Percentage change in the average power output of the wave energy converter with variation in heave plate mass from 900 kg to 800 kg across a range of sea states for design D5.

volume, a larger unstretched length and hence reduced pretension in the mooring.

F. Variation of average power between different designs

The average power obtained by the six different designs were compared for the PTO parameter ($k = 5000$ N/m and $B = 2000$ Ns/m). Though, these are not the optimal PTO parameters for the different designs, Table V provides an insight into the performance of various designs.

Table V
COMPARISON OF AVG POWER BETWEEN DIFFERENT DESIGNS

Design	$2R_f$ (m)	$2R_{hp}$ (m)	Average Power (W)
D0	2	2	286
D1	1.554	3	56.4
D2	1.64	2.96	101
D3	3	2.7	195
D4	5.3	2.4	146
D5	2.96	2.7	169
D6	3.5	2.7	172

G. Optimization of PTO parameters

A thorough search for optimal coefficients is done for all the designs. The average annual power is obtained by simulating 18 sea states. Each sea state was simulated using 58 different combinations of PTO coefficients (stiffness k and damping B coefficient). A contour plot showing the different values of power generated corresponding to the k and B coefficients for design D3 in shown in Fig. 12. The maximum power captured is 380 W at $k = 5000$ N/m and $B = 4000$ Ns/m. The red line in the contour plot represents the acceptable minimum power from the design based on the specification. Scaling the power with the occurrence of the sea state gives the average annual power for a particular site. This is shown in Fig. 13.

The values of the mechanical power generated at various sites and the corresponding PTO coefficients are tabulated in Table VI. Two methodologies were used to obtain the optimal power. These methods are referred to as PTO1 and PTO2 in Table VI.

Table VI
AVERAGE MECHANICAL POWER FOR DIFFERENT DESIGNS, PTO CONFIGURATIONS AND SITES. POWER UNIT IN WATTS (W)

Design		Deployment site					
		Oahu	Pac-Wave	San Diego	New Orleans	Cana-veral	Cape Cod
D0	PTO1	295	320	89	235	199	284
	PTO2	311	338	98	244	291	308
D2	PTO1	135	139	45	112	94	125
	PTO2	151	155	50	124	106	144
D3	PTO1	215	223	70	180	152	214
	PTO2	219	230	89	219	173	235
D6	PTO1	421	445	139	357	300	423
	PTO2	428	454	147	372	315	437

PTO1-Constant PTO coefficients for all sea states
PTO2-Optimal PTO coefficients for each sea state

- In PTO 1 the absolute power obtained by using a set of PTO coefficients at a particular sea state at that site is multiplied by the occurrence of that sea state at that site. This method is repeated for all the other PTO coefficients. The sum of the maximum power obtained by using a constant PTO is added to obtain the annual avg power.
- In PTO2, it is assumed that the PTO adapts to a combination of the coefficients, so that it produces max power at that sea state.

The maximum power is obtained at PacWave followed by Cape Code. The power obtained using an adaptive PTO (PTO2) is higher compared to the static PTO (PTO1) among all cases and sites. Approximately $8\% - 9\%$ improvement was recorded between the power generated using PTO2. This is expected as the system adapts to the sea state to deliver maximum power. The power obtained with D0 is also higher, compared to any other design. However, due to high tension in the tether, this design might incur higher material cost. D3 seems to be the ideal configuration as it reduces the material cost as well as delivers the required mechanical power. D6 also seems to be a good configuration as it delivers higher power compared to D3, though the material cost is expected to be higher due to the larger float diameter, it is also expected to accommodate all the components of the PTO system.

When comparing all the sites, D0 design produces a max annual avg power of 422 W followed by D5 at 355 W and D3 design which captures 223 W at a depth of 2 m. The red line in the contour plot represents the acceptable minimum power from the design based on the specification. Based on the contour plots the PTO parameters can be tuned to generate the maximum power for a particular design. Compared to D3, D0 and D5 show a narrow band of optimal PTO coefficient for maximum average power capture.

VIII. DISCUSSION AND CONCLUSION

This study investigated the hydrodynamic performance and design optimization of a submerged wave energy converter (WEC) equipped with a tether–PTO system. The paper first introduced the overall system architecture, including the prime movers, tether specifications, PTO algorithm, and PTO parameters. A numerical modelling framework was then developed, covering the governing theory, model implementation, verification of the developed UDF and the selection of representative sea states for performance assessment. The mechanical average annual power capture at six different sites along the US coasts for different designs is compared.

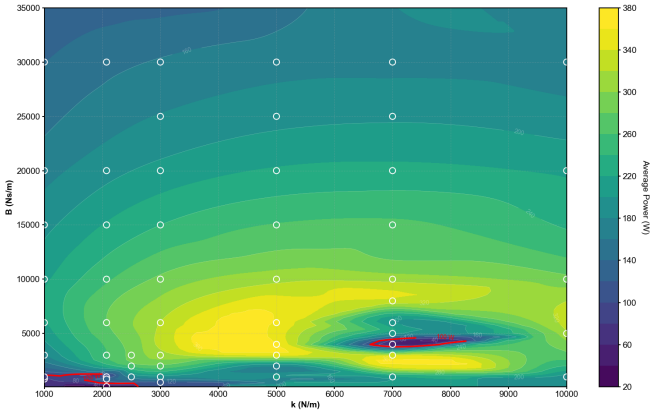


Figure 12. Absolute power generated for various PTO coefficients for design D3

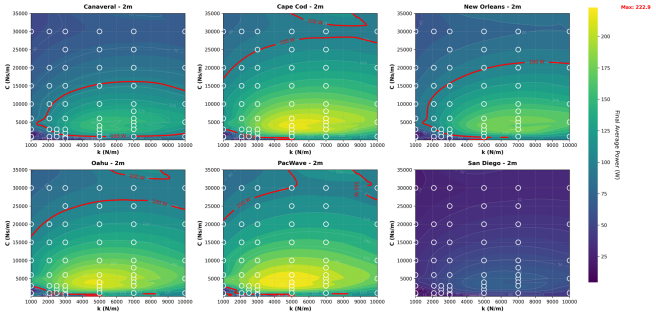


Figure 13. Optimal contour for PTO coefficients for design D3at various sites

A comprehensive sensitivity analysis was conducted under both survival and operational sea states. The operational state analysis examined the influence of key design parameters for six different designs of float and heave plate. These designs vary in characteristic dimensions. Several sensitivity analyses were carried out for different operating parameters, primarily the PTO coefficients, draft, initial tether length, mass ratio between float and heave plate and mass of heave plate. The results demonstrate that the hydrodynamic response and power capture of the submerged WEC are strongly influenced by both the system configuration and the PTO parameters. The sensitivity analyses showed that careful selection of key design parameters, including the geometric configuration, mass distribution, and the initial float–heave plate gap, can substantially enhance energy capture while maintaining acceptable structural loads and system stability. For a configuration that is submerged 2 m below the surface, the power capture at most of the sites for most of the PTO coefficients is more than the minimum targeted

value (only location where power was lower was in San Diego). The variation of power capture at various depths and different tether lengths is also compared for two designs. A linear drop in the captured power is observed with an increase in the draft and increase in the initial tether length (over the range of 2-7 m submerged). The optimal PTO coefficient at six different sites around US were found out for maximum power capture.

Overall, the validated numerical model and the comprehensive parametric analyses presented in this study provide a robust framework for optimizing submerged WEC designs, enabling an effective balance between energy production, structural integrity, operational reliability and survivability for site-specific deployment.

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